GRASSWORKS: Analysis of Bauer et al. (submitted) Functional traits of grasslands: Community weighted mean of canopy height per plot (esy4 = 4qm plots)

MB

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> **NOTE:** To compare different models, you only have to change the models in the section 'Load models'

Preparation

Protocol of data exploration (Steps 1-8) used from Zuur et al. (2010) Methods Ecol Evol DOI: 10.1111/2041-210X.12577

```r
library(here)
library(tidyverse)
```

Packages

```
## Warning: Paket 'ggplot2' wurde unter R Version 4.5.3 erstellt
## Warning: Paket 'purrr' wurde unter R Version 4.5.3 erstellt
## Warning: Paket 'dplyr' wurde unter R Version 4.5.3 erstellt
```

```
library(ggbeeswarm)
library(patchwork)
library(DHARMa)
library(emmeans)
```

```
sites <- read_csv(
  here("data", "processed", "data_processed_sites.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(
    .default = "?",
    eco.id = "f",
    region = col_factor(levels = c("north", "centre", "south"), ordered = TRUE),
    site.type = col_factor(
      levels = c("positive", "restored", "negative"), ordered = TRUE
      ),
    obs.year = "f",
    hydrology = "f",
    mngm.type = "f"
  )
) %>%
  mutate(
    esy4 = fct_relevel(esy4, "R22", "R1A"),
    eco.id = factor(eco.id)
    ) %>%
  rename(y = cwm.abu.height) %>%
  filter(y < 1) # see section Outliers: Exclude site N_DAM (more or less only the tall grass Arrhenathe
```

Load data

Data exploration

Means and deviations

```
Rmisc::CI(sites$y, ci = .95)
```

```
##     upper      mean     lower
## 0.4634648 0.4485813 0.4336978
```

```
median(sites$y)
```

```
## [1] 0.44
```

```
sd(sites$y)
```

```
## [1] 0.1285512
```

```
quantile(sites$y, probs = c(0.05, 0.95), na.rm = TRUE)
```

```
##    5%   95%
## 0.228 0.686
```

```r
sites %>% count(eco.id)
```

```
## # A tibble: 3 x 2
##   eco.id     n
##   <fct>  <int>
## 1 654      101
## 2 664       89
## 3 686       99
```

```r
sites %>% count(site.type)
```

```
## # A tibble: 3 x 2
##   site.type     n
##   <ord>     <int>
## 1 positive     45
## 2 restored    219
## 3 negative     25
```

```r
sites %>% count(esy4)
```

```
## # A tibble: 2 x 2
##   esy4      n
##   <fct> <int>
## 1 R22     208
## 2 R1A      81
```

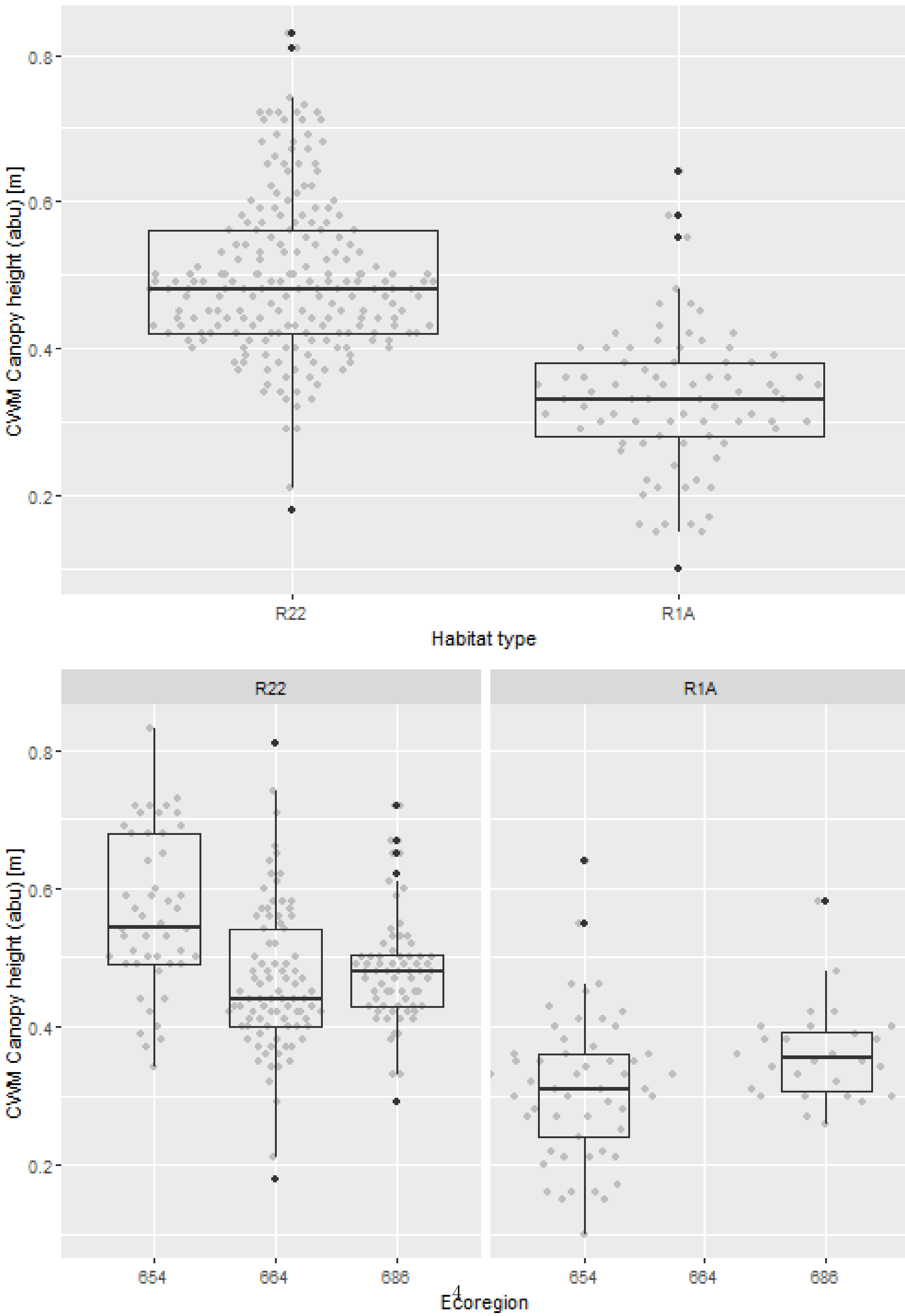
```r
sites %>% count(esy4, eco.id)
```

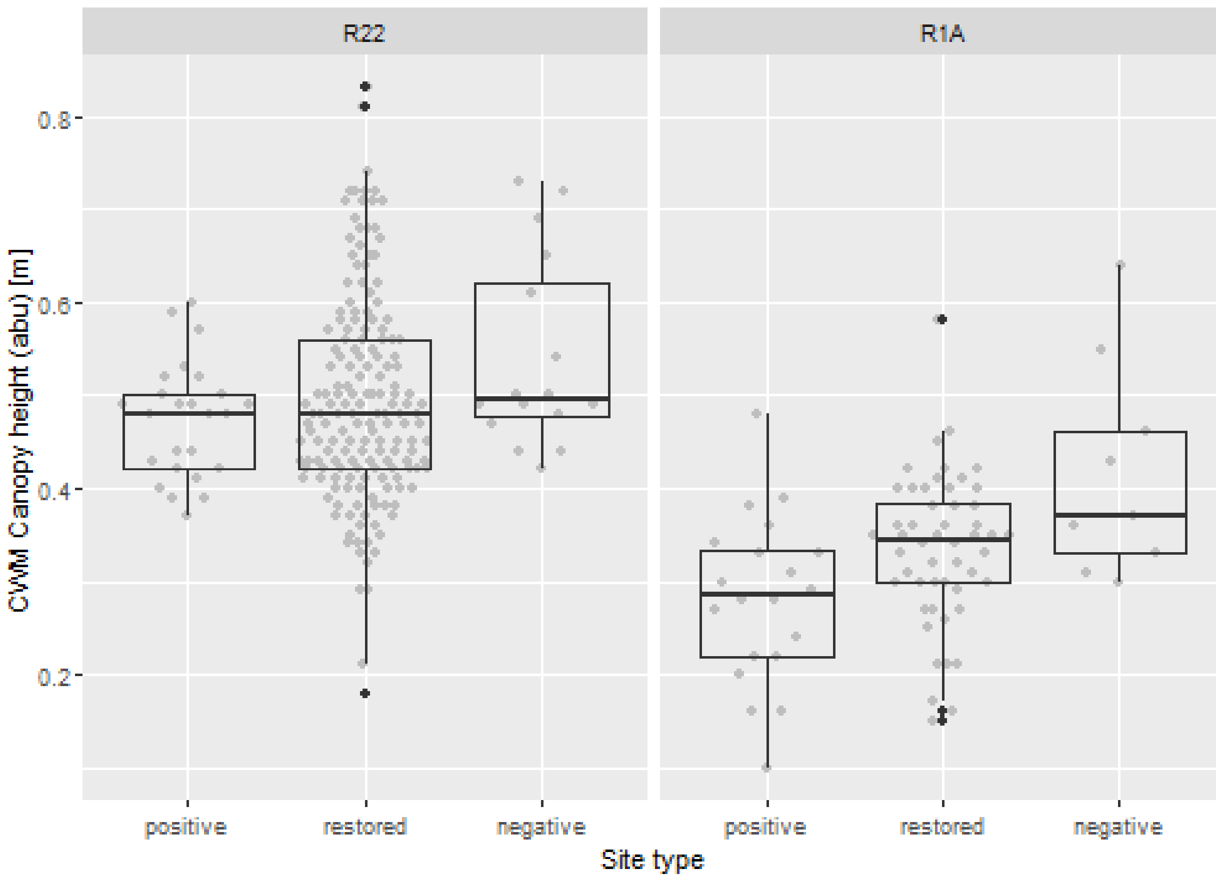
```
## # A tibble: 5 x 3
##   esy4  eco.id     n
##   <fct> <fct>  <int>
## 1 R22   654       48
## 2 R22   664       89
## 3 R22   686       71
## 4 R1A   654       53
## 5 R1A   686       28
```

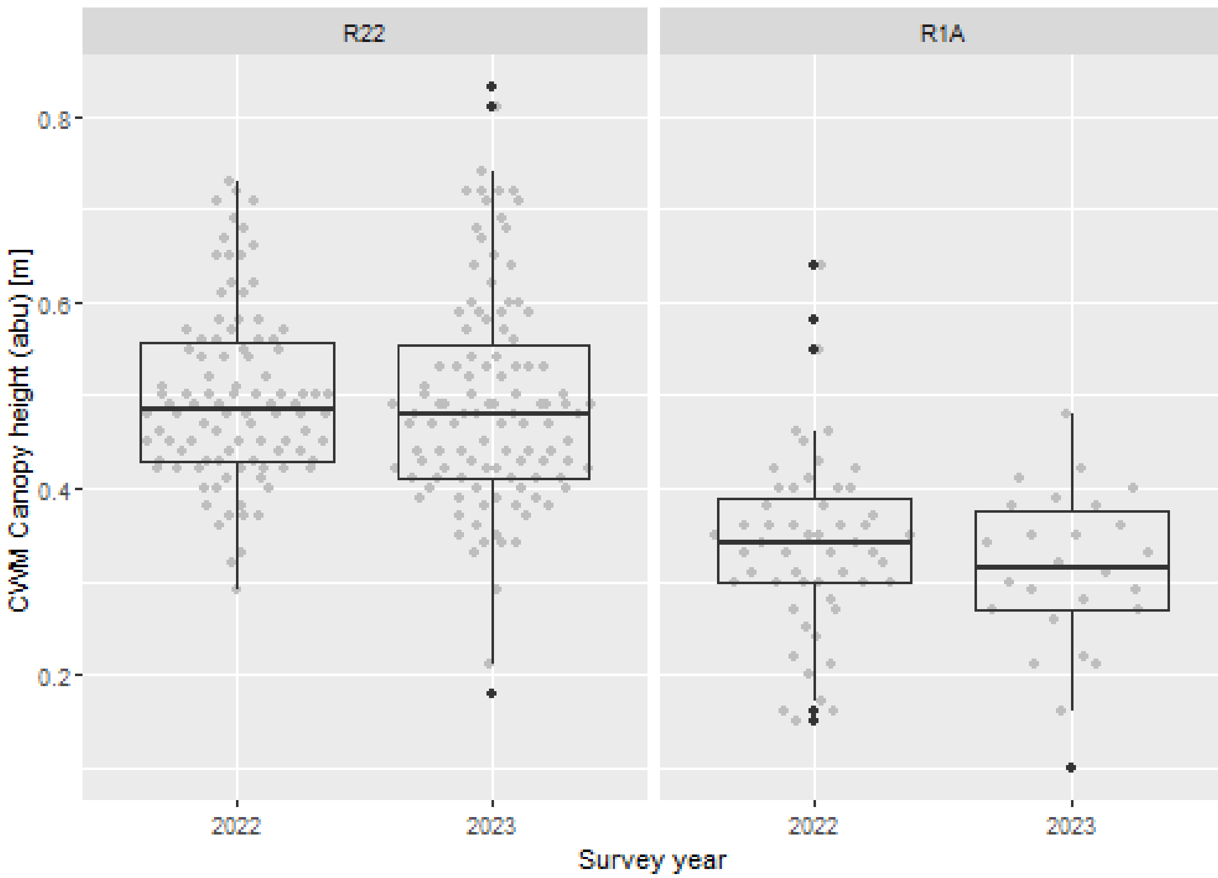
```r
sites %>% count(esy4, site.type)
```

```
## # A tibble: 6 x 3
##   esy4  site.type     n
##   <fct> <ord>     <int>
## 1 R22   positive     25
## 2 R22   restored    167
## 3 R22   negative     16
## 4 R1A   positive     20
## 5 R1A   restored     52
## 6 R1A   negative      9
```

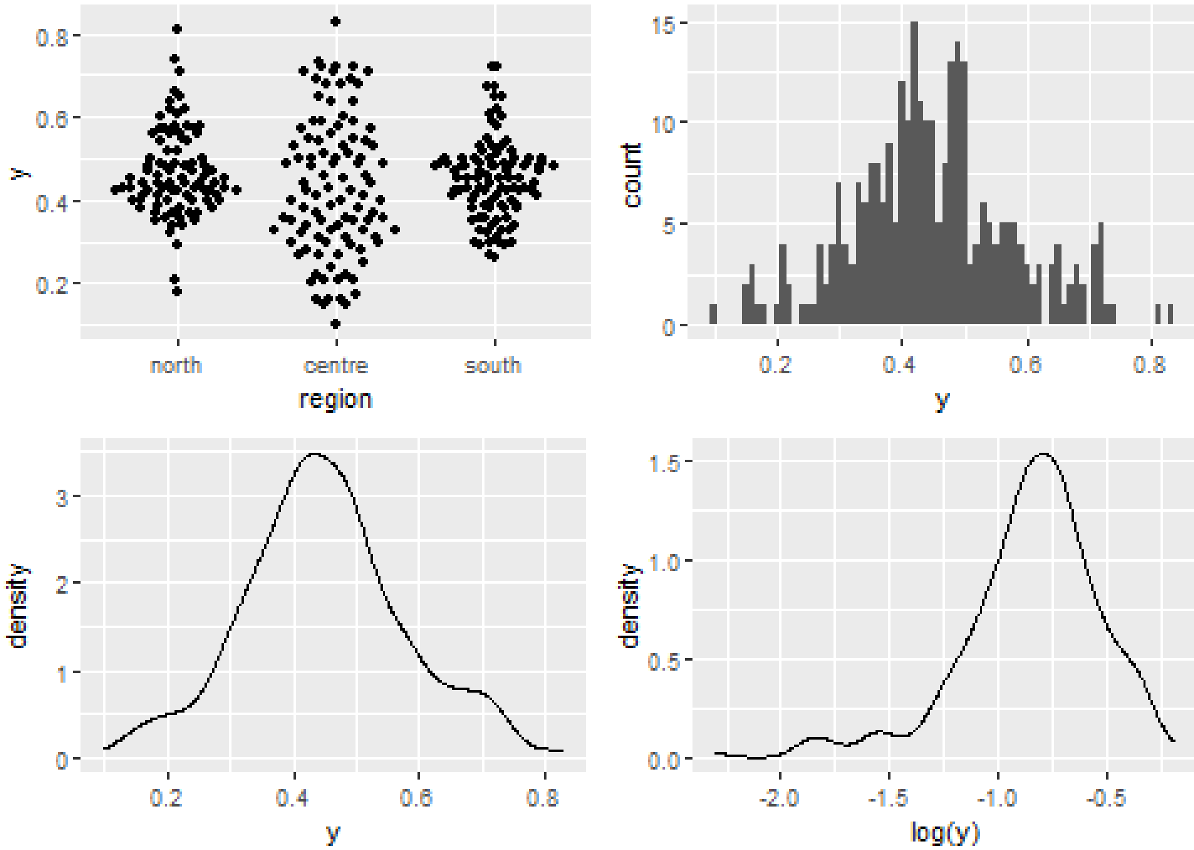
Graphs of raw data (Step 2, 6, 7)




R22
R1A
CWM Canopy height (abu) [m]
0.8
0.6
0.4
0.2
positive
restored
negative
positive
restored
negative
Site type


R22
R1A
0.8
0.6
0.4
0.2
CWM Canopy height (abu) [m]
2022
2023
2022
2023
Survey year

Outliers, zero-inflation, transformations? (Step 1, 3, 4)



Check collinearity part 1 (Step 5)

Exclude r > 0.7 Dormann et al. 2013 Ecography DOI: 10.1111/j.1600-0587.2012.07348.x

```
# sites %>%
#   select(where(is.numeric), -y, -starts_with("cwm.")) %>%
#   GGally::ggpairs(
#     lower = list(continuous = "smooth_loess")
#     ) +
#   theme(strip.text = element_text(size = 7))

# -> no continuous variables
```

Models

NOTE: Only here you have to modify the script to compare other models

```
load(file = here("outputs", "models", "model_height_cwm_1.Rdata"))
load(file = here("outputs", "models", "model_height_cwm_2.Rdata"))
m_1 <- m1
m_2 <- m2
```

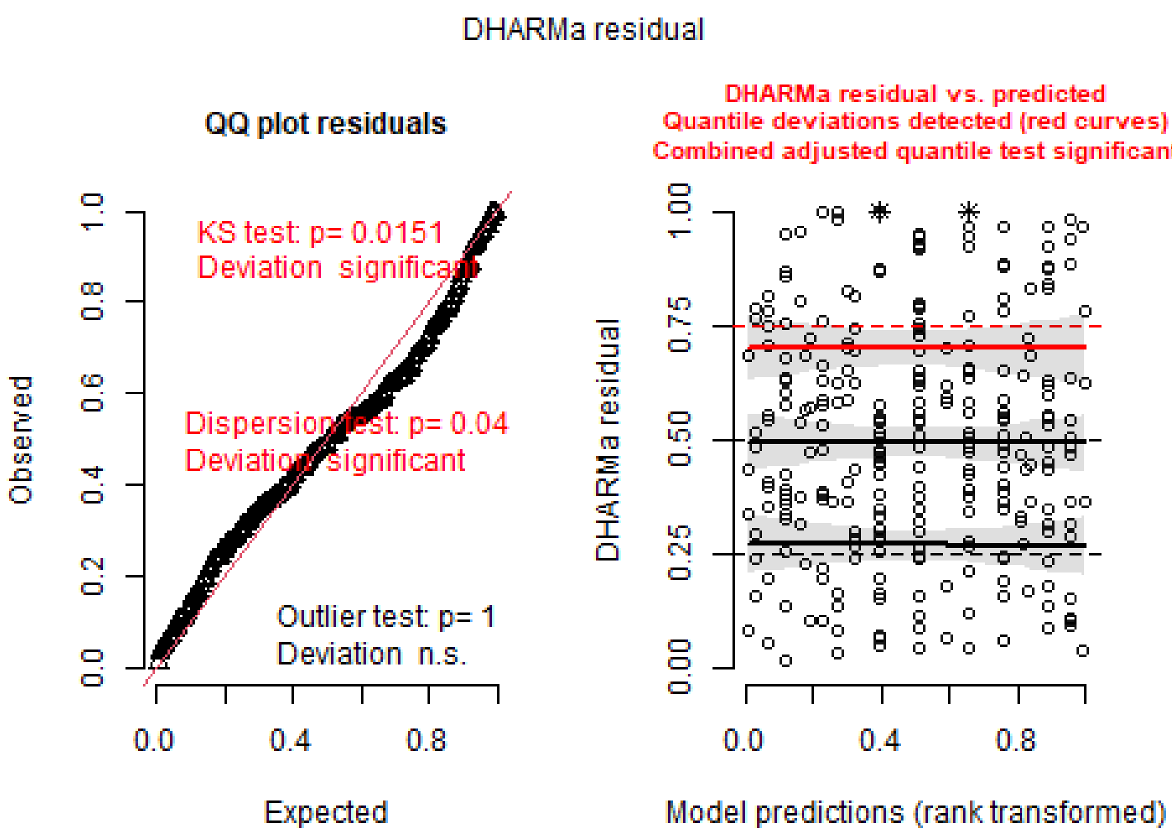
```
m_1@call
## lmer(formula = y ~ esy4 * (site.type + eco.id) + obs.year + (1 |
```

```
##     id.site), data = sites, REML = FALSE)
m_2@call
## lmer(formula = y ~ (esy4 + site.type + eco.id + obs.year)^2 +
##     (1 | id.site), data = sites, REML = FALSE)
```

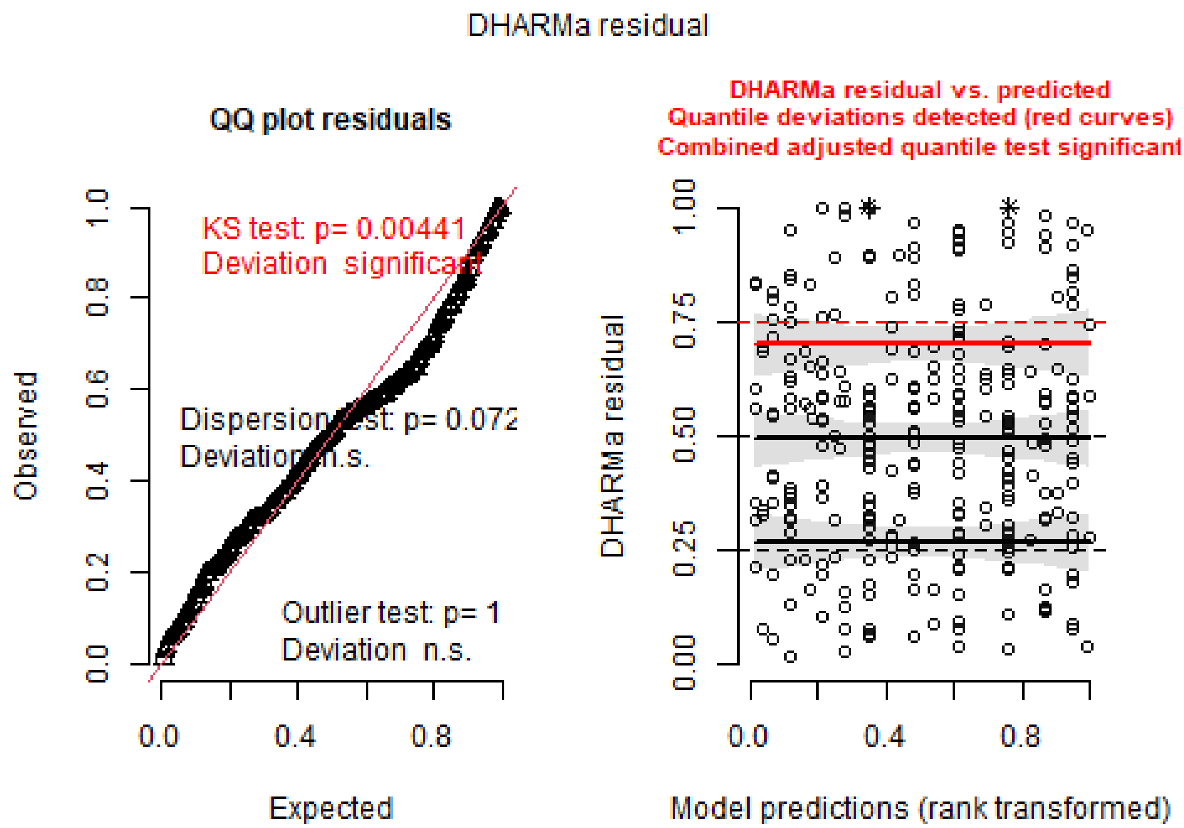
Model check

DHARMa

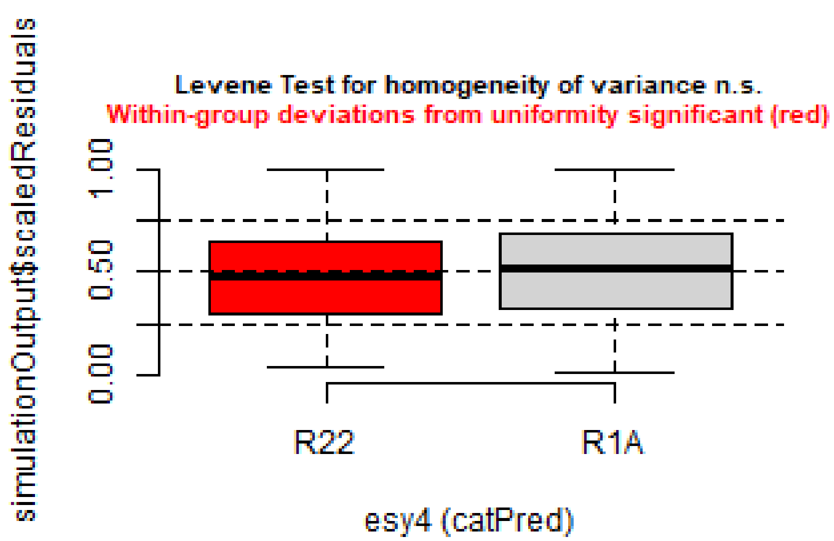
```
simulation_output_1 <- simulateResiduals(m_1, plot = TRUE)
```



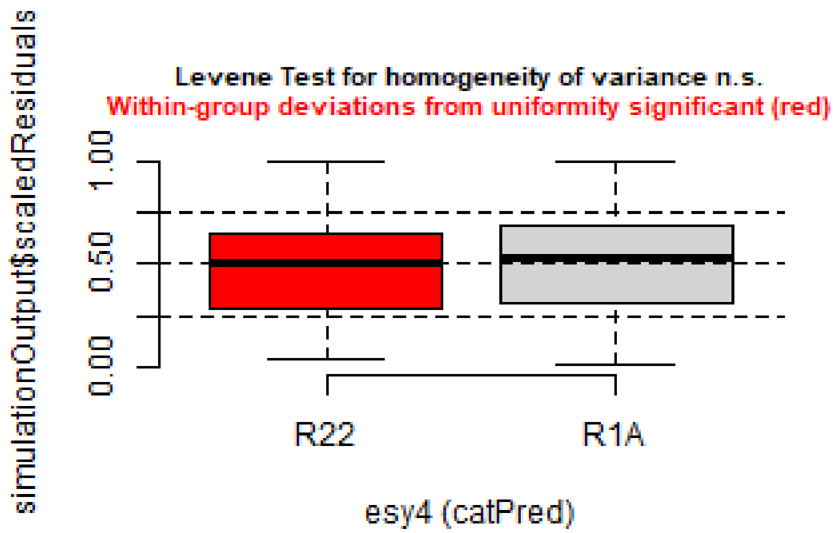

```
simulation_output_2 <- simulateResiduals(m_2, plot = TRUE)
```



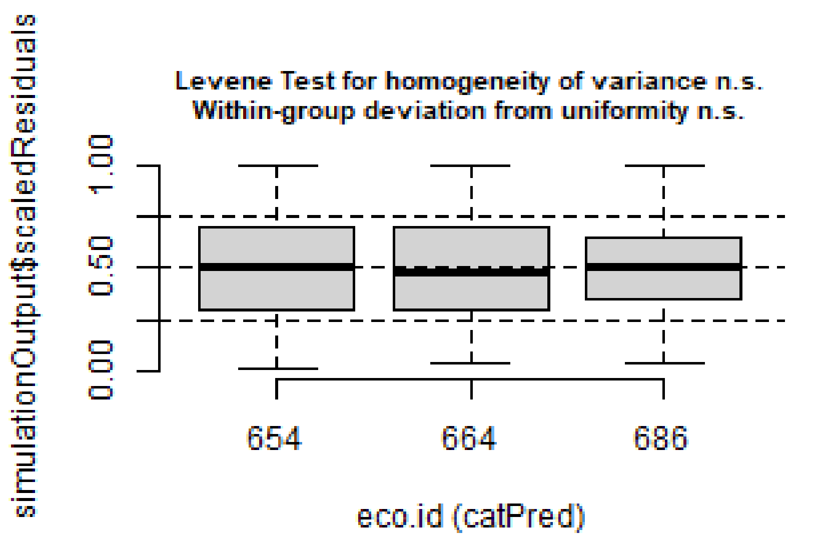

```
plotResiduals(simulation_output_1$scaledResiduals, sites$esy4)
```



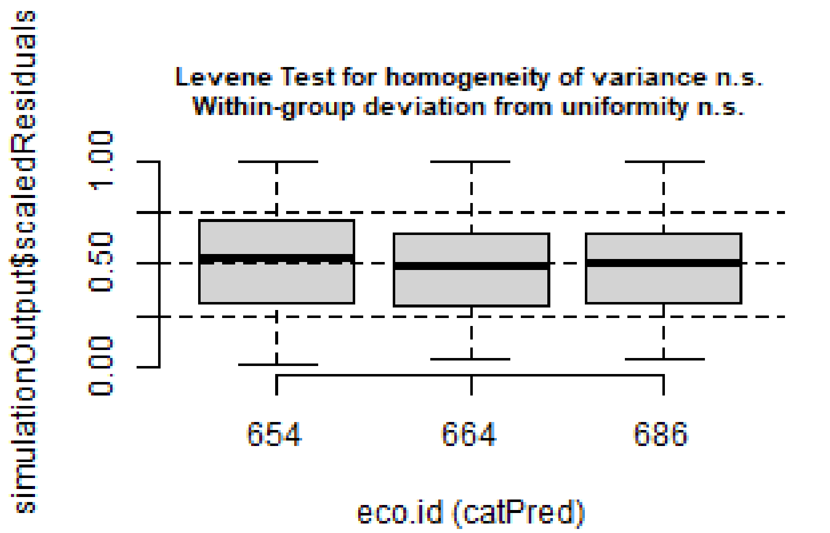

```
plotResiduals(simulation_output_2$scaledResiduals, sites$esy4)
```



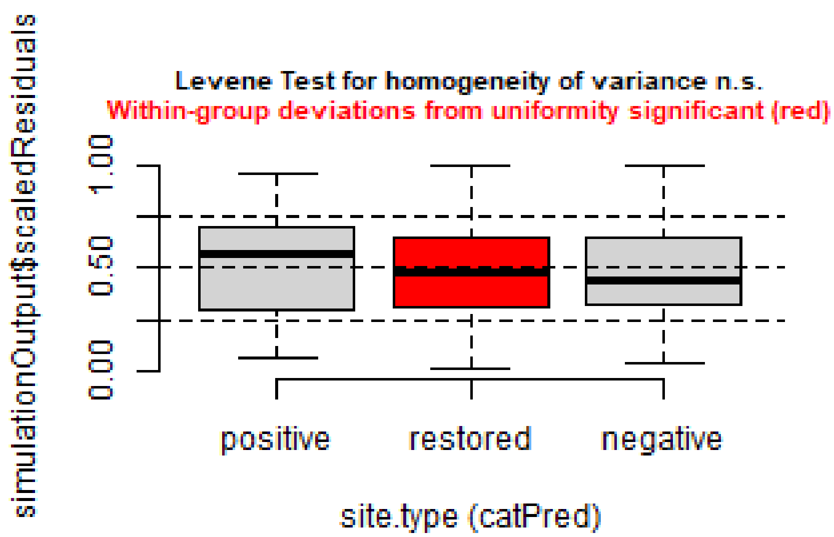

```r
plotResiduals(simulation_output_1$scaledResiduals, sites$eco.id)
```



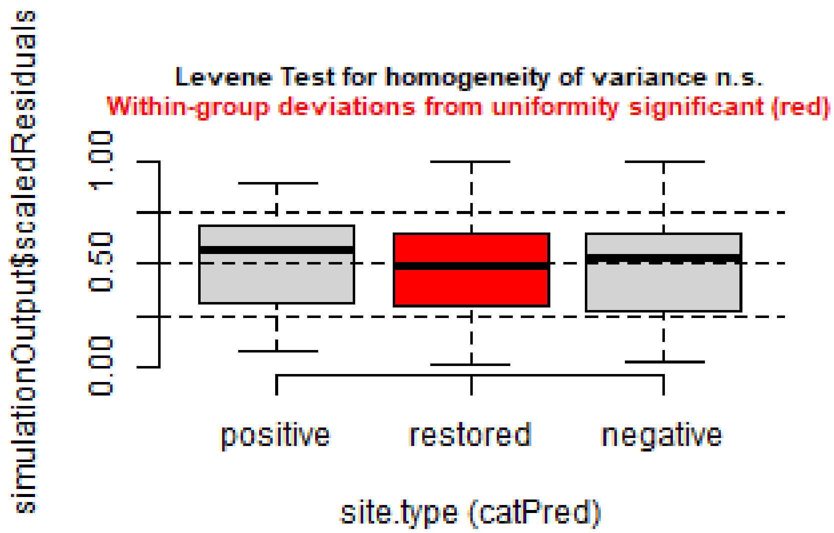

```r
plotResiduals(simulation_output_2$scaledResiduals, sites$eco.id)
```



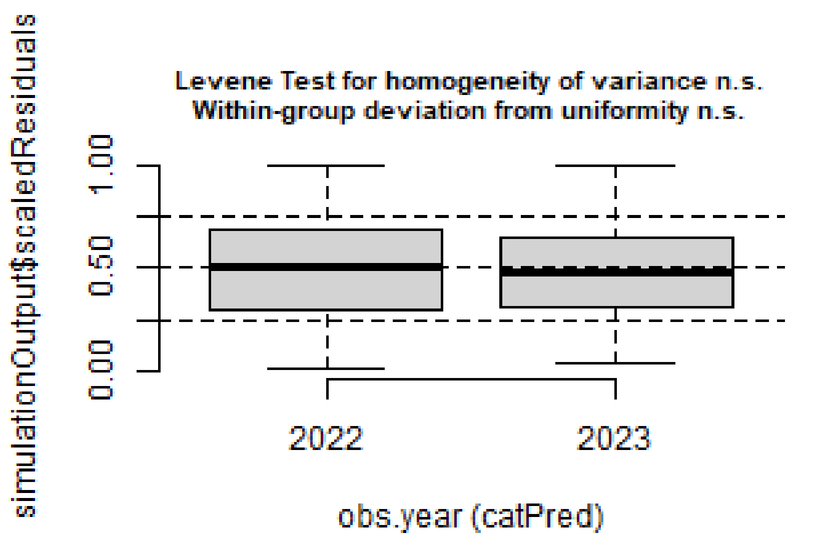

```r
plotResiduals(simulation_output_1$scaledResiduals, sites$site.type)
```



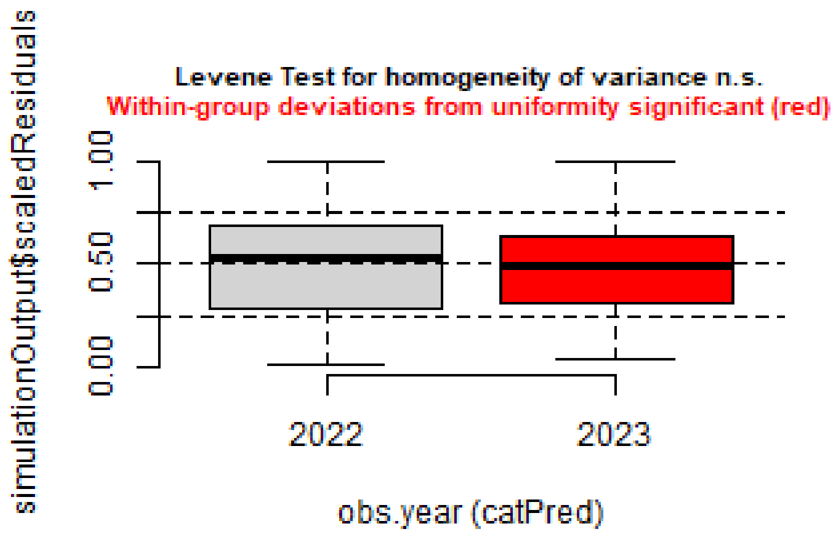

```r
plotResiduals(simulation_output_2$scaledResiduals, sites$site.type)
```



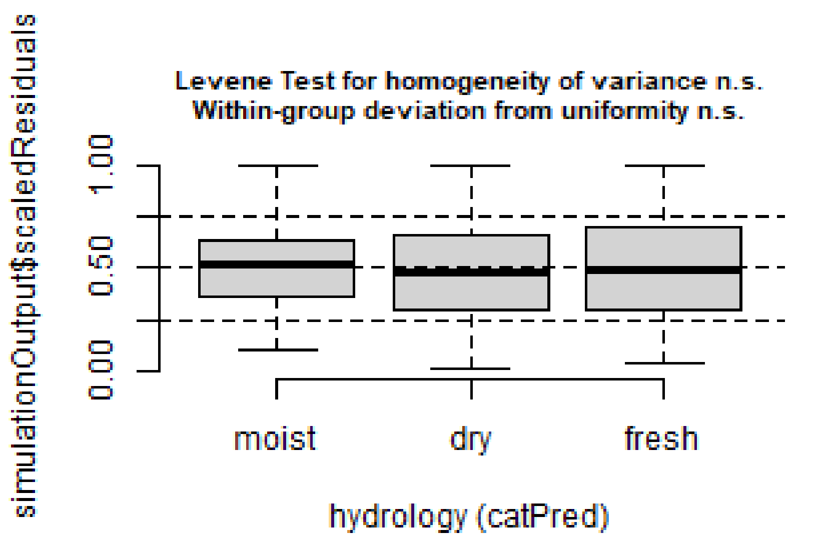

```r
plotResiduals(simulation_output_1$scaledResiduals, sites$obs.year)
```



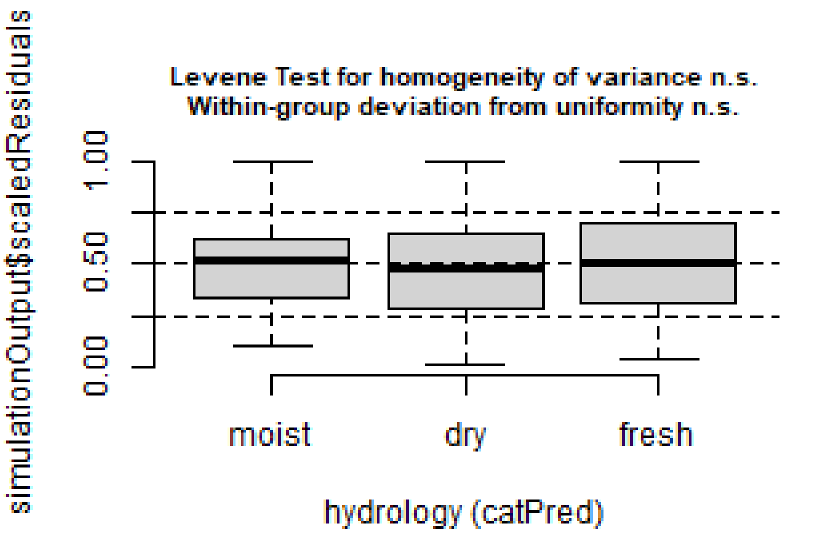

```r
plotResiduals(simulation_output_2$scaledResiduals, sites$obs.year)
```



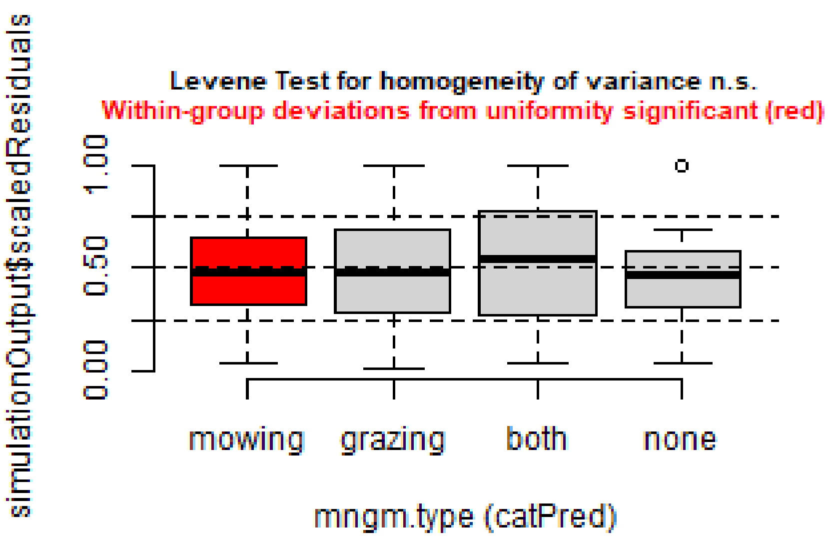

```r
plotResiduals(simulation_output_1$scaledResiduals, sites$hydrology)
```



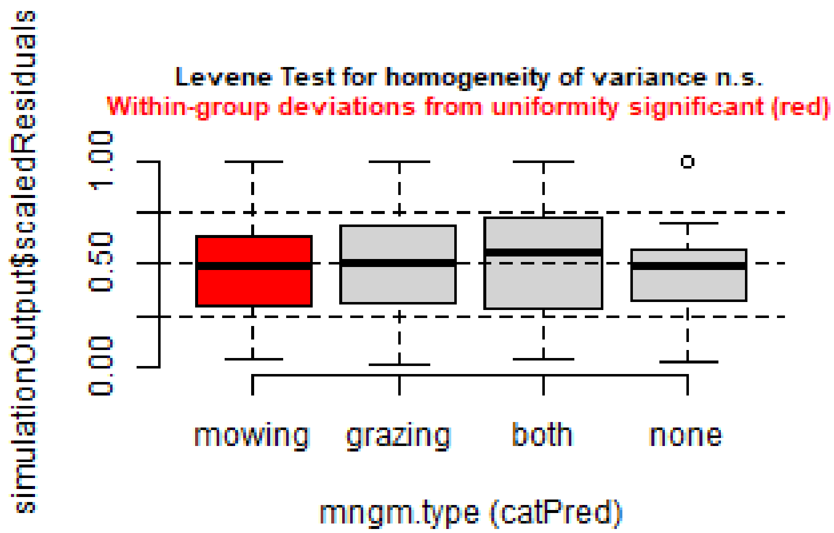

```r
plotResiduals(simulation_output_2$scaledResiduals, sites$hydrology)
```




```r
plotResiduals(simulation_output_1$scaledResiduals, sites$mngm.type)
```




```r
plotResiduals(simulation_output_2$scaledResiduals, sites$mngm.type)
```



Check collinearity part 2 (Step 5)

Remove VIF > 3 or > 10 Zuur et al. 2010 Methods Ecol Evol DOI: 10.1111/j.2041-210X.2009.00001.x

```
car::vif(m_1)
```

```
## Registered S3 method overwritten by 'car':
##   method          from
##   na.action.merMod lme4

##                     GVIF Df GVIF^(1/(2*Df))
## esy4            3.025606  1        1.739427
## site.type       1.976839  2        1.185749
## eco.id          1.769052  2        1.153281
## obs.year        1.046940  1        1.023201
## esy4:site.type  3.250758  2        1.342753
## esy4:eco.id     2.248951  1        1.499650
```

```
car::vif(m_2)
```

```
##                          GVIF Df GVIF^(1/(2*Df))
## esy4                 5.540917  1        2.353915
## site.type           59.153304  2        2.773287
## eco.id              21.268885  2        2.147515
## obs.year             6.206095  1        2.491204
## esy4:site.type       9.398477  2        1.750912
## esy4:eco.id          2.585528  1        1.607958
## esy4:obs.year        2.998932  1        1.731743
## site.type:eco.id    77.800058  4        1.723346
## site.type:obs.year   9.600051  2        1.760226
## eco.id:obs.year     11.673695  2        1.848426
```

Model comparison

R2 values

```r
MuMIn::r.squaredGLMM(m_1)
##            R2m      R2c
## [1,] 0.4169676 0.715171
MuMIn::r.squaredGLMM(m_2)
##            R2m       R2c
## [1,] 0.4546629 0.7235414
```

AICc

Use AICc and not AIC since ratio n/K < 40 Burnahm & Anderson 2002 p. 66 ISBN: 978-0-387-95364-9

```r
MuMIn::AICc(m_1, m_2) %>%
  arrange(AICc)
##     df      AICc
## m_1 12 -580.7186
## m_2 21 -570.2147
```

Predicted values

Summary table

```r
car::Anova(m_1, type = 3)
```

```
## Analysis of Deviance Table (Type III Wald chisquare tests)
##
## Response: y
##                   Chisq Df Pr(>Chisq)
## (Intercept)    692.5248  1  < 2.2e-16 ***
## esy4            68.3004  1  < 2.2e-16 ***
## site.type        2.6087  2  0.2713428
## eco.id          15.6626  2  0.0003971 ***
## obs.year         0.2986  1  0.5847781
## esy4:site.type   1.5111  2  0.4697597
## esy4:eco.id     13.2930  1  0.0002664 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
summary(m_1)
```

```
## Linear mixed model fit by maximum likelihood  ['lmerMod']
## Formula: y ~ esy4 * (site.type + eco.id) + obs.year + (1 | id.site)
##    Data: sites
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    -581.8    -537.9     302.9    -605.8       277
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.08092 -0.48018 -0.01625  0.41035  3.04610
##
```

```
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  id.site  (Intercept) 0.004589 0.06774
##  Residual             0.004383 0.06620
## Number of obs: 289, groups:  id.site, 118
##
## Fixed effects:
##                       Estimate Std. Error t value
## (Intercept)           0.565453   0.021487  26.316
## esy4R1A              -0.230425   0.027882  -8.264
## site.type.L           0.038627   0.025364   1.523
## site.type.Q           0.012921   0.016537   0.781
## eco.id664            -0.087607   0.022400  -3.911
## eco.id686            -0.062701   0.022091  -2.838
## obs.year2023         -0.008427   0.015422  -0.546
## esy4R1A:site.type.L   0.055598   0.045928   1.211
## esy4R1A:site.type.Q   0.008598   0.030999   0.277
## esy4R1A:eco.id686     0.130702   0.035848   3.646
##
## Correlation of Fixed Effects:
##             (Intr) es4R1A st.t.L st.t.Q ec.664 ec.686 o.2023 e4R1A:..L
## esy4R1A     -0.641
## site.type.L -0.010 -0.042
## site.type.Q  0.447 -0.324  0.165
## eco.id664   -0.671  0.469  0.131 -0.063
## eco.id686   -0.632  0.458  0.056 -0.025  0.602
## obs.yer2023 -0.413  0.093  0.114 -0.018  0.073  0.003
## esy4R1A:..L -0.063  0.253 -0.511 -0.122 -0.041 -0.010  0.019
## esy4R1A:..Q -0.227  0.480 -0.119 -0.493  0.036  0.022  0.002  0.391
## es4R1A:.686  0.347 -0.506 -0.043  0.014 -0.341 -0.537  0.026  0.170
##             e4R1A:..Q
## esy4R1A
## site.type.L
## site.type.Q
## eco.id664
## eco.id686
## obs.yer2023
## esy4R1A:..L
## esy4R1A:..Q
## es4R1A:.686  0.136
## fit warnings:
## fixed-effect model matrix is rank deficient so dropping 1 column / coefficient
```

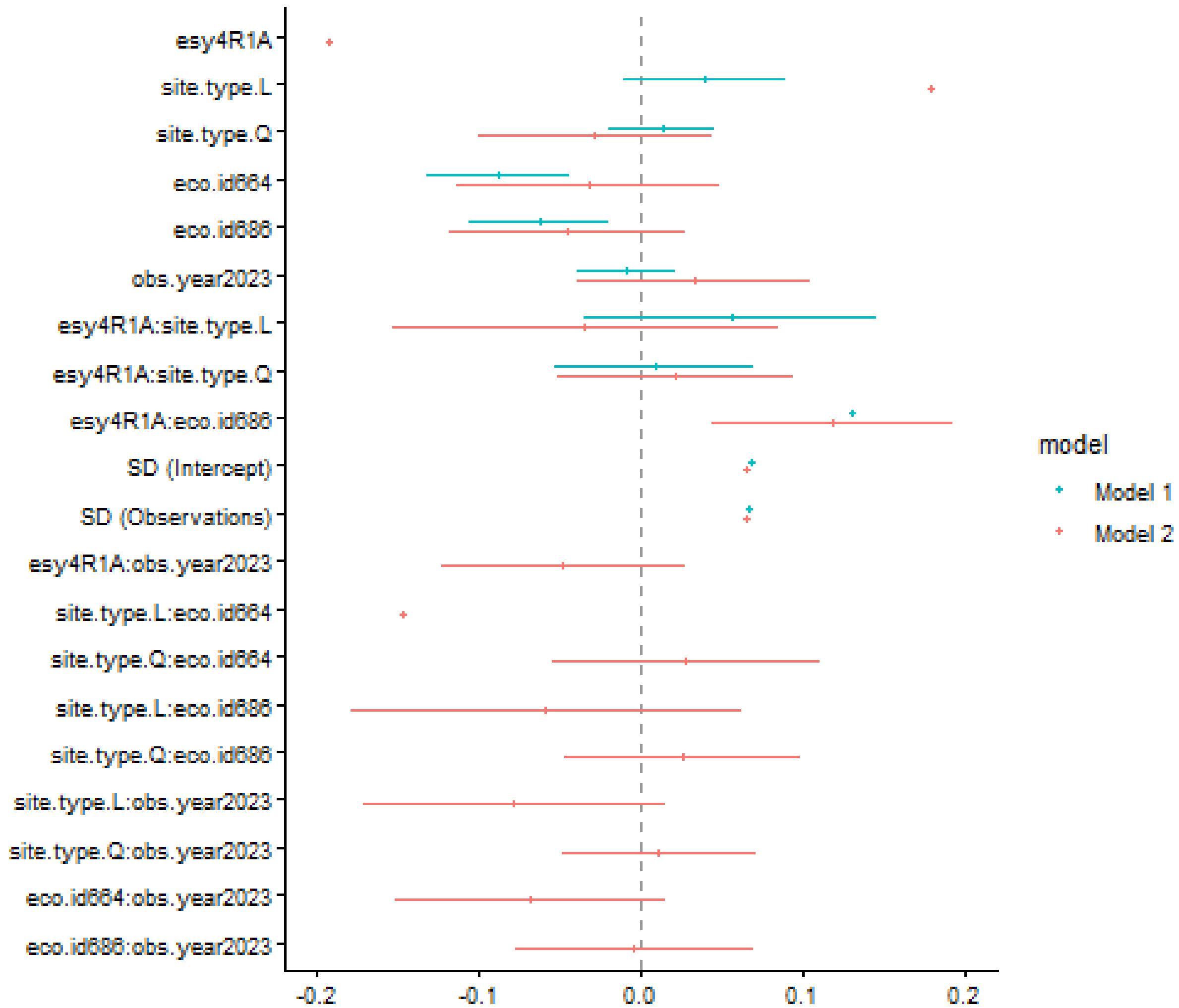
Forest plot

```r
dotwhisker::dwplot(
  list(m_1, m_2),
  ci = 0.95,
  show_intercept = FALSE,
  vline = geom_vline(xintercept = 0, colour = "grey60", linetype = 2)) +
  xlim(-0.2, 0.2) +
  theme_classic()
```

```
## Package 'merDeriv' needs to be installed to compute confidence intervals
```

```
##   for random effect parameters.
## Package 'merDeriv' needs to be installed to compute confidence intervals
##   for random effect parameters.
```

```
## Warning: Using the `size` aesthetic with geom_segment was deprecated in ggplot2 3.4.0.
## i Please use the `linewidth` aesthetic instead.
## i The deprecated feature was likely used in the dotwhisker package.
##   Please report the issue at <https://github.com/fsolt/dotwhisker/issues>.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



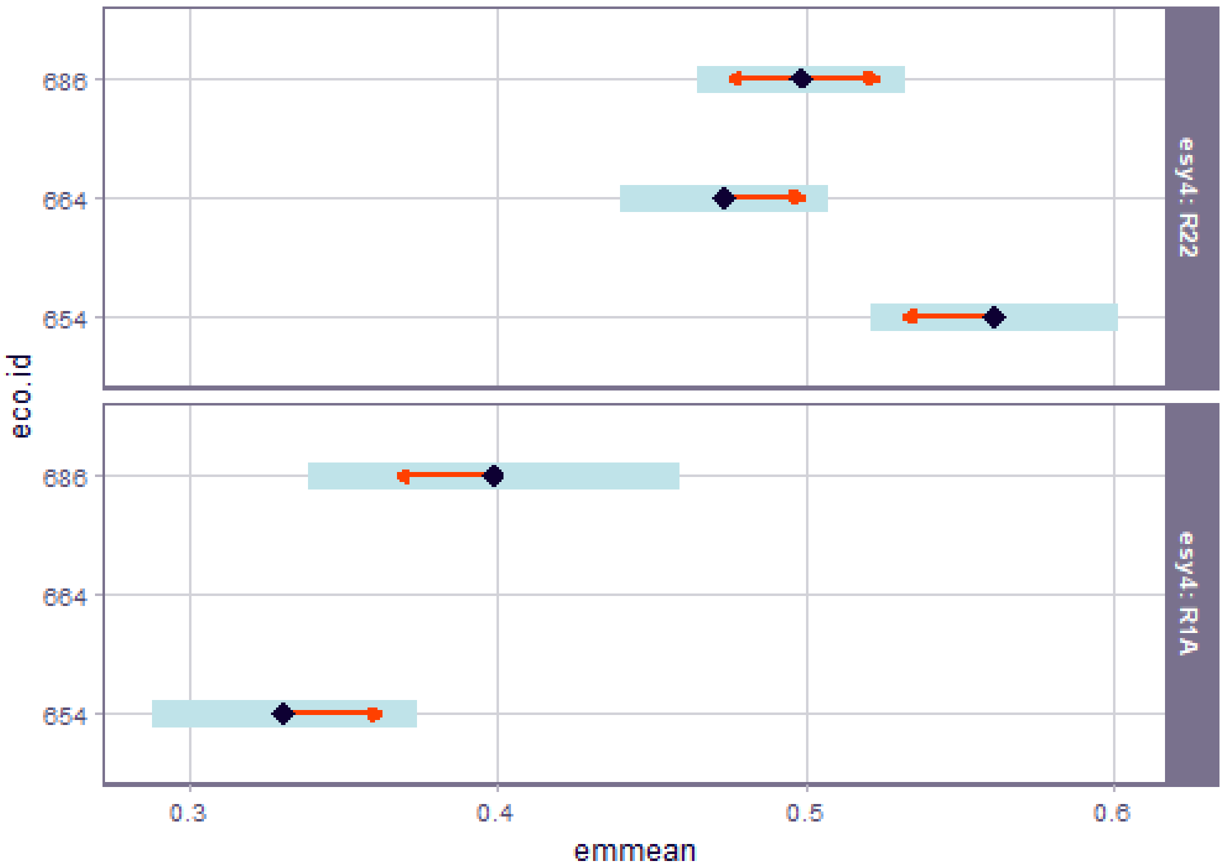
Effect sizes: Habiat type x Region

```
emm <- emmeans(
  m_1,
  revpairwise ~ eco.id | esy4,
  type = "response"
  )
plot(emm, comparison = TRUE)
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_segment()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```




```
emm
```

```
## $emmeans
## esy4 = R22:
##  eco.id emmean     SE  df lower.CL upper.CL
##  654     0.561 0.0204 160    0.521    0.601
##  664     0.474 0.0170 130    0.440    0.507
##  686     0.499 0.0172 162    0.465    0.532
##
## esy4 = R1A:
##  eco.id emmean     SE  df lower.CL upper.CL
##  654     0.331 0.0217 129    0.288    0.374
##  664    nonEst     NA  NA       NA       NA
##  686     0.399 0.0305 158    0.339    0.459
##
## Results are averaged over the levels of: site.type, obs.year
## Degrees-of-freedom method: kenward-roger
```

```
## Confidence level used: 0.95
##
## $contrasts
## esy4 = R22:
##  contrast              estimate     SE  df t.ratio p.value
##  eco.id664 - eco.id654  -0.0876 0.0233 140  -3.756  0.0007
##  eco.id686 - eco.id654  -0.0627 0.0230 154  -2.730  0.0193
##  eco.id686 - eco.id664   0.0249 0.0207 137   1.205  0.4520
##
## esy4 = R1A:
##  contrast              estimate     SE  df t.ratio p.value
##  eco.id664 - eco.id654   nonEst     NA  NA      NA      NA
##  eco.id686 - eco.id654   0.0680 0.0316 165   2.149  0.0331
##  eco.id686 - eco.id664   nonEst     NA  NA      NA      NA
##
## Results are averaged over the levels of: site.type, obs.year
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for varying family sizes
```

Session info

```
## R version 4.5.2 (2025-10-31 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
##
## time zone: Europe/Berlin
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] emmeans_2.0.3    DHARMa_0.5.0     patchwork_1.3.2  ggbeeswarm_0.7.3
##  [5] lubridate_1.9.5  forcats_1.0.1    stringr_1.6.0    dplyr_1.2.1
##  [9] purrr_1.2.2      readr_2.2.0      tidyr_1.3.2      tibble_3.3.1
## [13] ggplot2_4.0.3    tidyverse_2.0.0  here_1.0.2
##
## loaded via a namespace (and not attached):
##  [1] Rdpack_2.6.6          gridExtra_2.3         sandwich_3.1-1
##  [4] rlang_1.2.0           magrittr_2.0.5        multcomp_1.4-30
##  [7] otel_0.2.0            compiler_4.5.2        mgcv_1.9-3
## [10] vctrs_0.7.3           pkgconfig_2.0.3       crayon_1.5.3
## [13] fastmap_1.2.0         backports_1.5.1       labeling_0.4.3
## [16] utf8_1.2.6            ggstance_0.3.7        promises_1.5.0
## [19] rmarkdown_2.31        tzdb_0.5.0            nloptr_2.2.1
```

```
## [22] bit_4.6.0               xfun_0.58             later_1.4.8
## [25] broom_1.0.13            parallel_4.5.2        R6_2.6.1
## [28] gap.datasets_0.0.6      stringi_1.8.7         qgam_2.0.0
## [31] RColorBrewer_1.1-3      car_3.1-5             boot_1.3-32
## [34] estimability_1.5.1      Rcpp_1.1.1-1.1        iterators_1.0.14
## [37] knitr_1.51              zoo_1.8-15            parameters_0.29.1
## [40] httpuv_1.6.17           Matrix_1.7-4          splines_4.5.2
## [43] timechange_0.4.0        tidyselect_1.2.1      rstudioapi_0.18.0
## [46] abind_1.4-8             yaml_2.3.12           MuMIn_1.48.19
## [49] doParallel_1.0.17       codetools_0.2-20      lattice_0.22-7
## [52] plyr_1.8.9              bayestestR_0.18.1     shiny_1.13.0
## [55] withr_3.0.2             S7_0.2.2              evaluate_1.0.5
## [58] marginaleffects_0.32.0 survival_3.8-3        pillar_1.11.1
## [61] gap_1.15.2              carData_3.0-6         foreach_1.5.2
## [64] stats4_4.5.2            insight_1.5.1         reformulas_0.4.4
## [67] generics_0.1.4          vroom_1.7.1           rprojroot_2.1.1
## [70] hms_1.1.4               scales_1.4.0          minqa_1.2.8
## [73] xtable_1.8-8            glue_1.8.1            tools_4.5.2
## [76] data.table_1.18.4       lme4_2.0-1            mvtnorm_1.4-0
## [79] grid_4.5.2              rbibutils_2.4.1       datawizard_1.3.1
## [82] nlme_3.1-168            Rmisc_1.5.1           performance_0.17.0
## [85] beeswarm_0.4.0          vipor_0.4.7           Formula_1.2-5
## [88] cli_3.6.6               gtable_0.3.6          digest_0.6.39
## [91] pbkrtest_0.5.5          TH.data_1.1-5         farver_2.1.2
## [94] htmltools_0.5.9         lifecycle_1.0.5       mime_0.13
## [97] bit64_4.8.2             dotwhisker_0.8.4      MASS_7.3-65
```